
CS683 PA-1

Task 2 Report

Matrix Multiplication for Inference (SGEMM)

Software Prefetching · SIMD · llama.cpp Integration

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Contents

Setup	2
1 Task 2A: Software Prefetching	2
1.1 Implementation	2
1.2 Baseline Profiling (perf)	2
1.3 Speedup vs. Matrix Size	2
1.4 Speedup vs. Prefetch Distance	3
1.5 Speedup vs. Prefetch Locality Hint	3
1.6 Effect of Hardware Prefetchers	3
1.7 Questions	3
2 Task 2B: SIMD (AVX2)	4
2.1 Implementation	4
2.2 Baseline Profiling / Instruction Count	4
2.3 Performance	5
2.4 Speedup vs. SIMD Width	5
2.5 Questions	7
3 Task 2C: Software Prefetching + SIMD (Combined / Optimized)	7
3.1 Implementation	7
3.2 Combinations Evaluated	7
3.3 Final Performance Summary	8
3.4 Harness Score	8
3.5 Discussion	8
4 llama.cpp Integration	8
4.1 Setup	8
4.2 Results	9
4.3 Discussion	9

Setup

All measurements were taken on a 13th Gen Intel Core i7-13650HX. Every build and every run was pinned to CPU 0, a P-core, with `taskset -c 0`. L1-D size is 48KB and line size is 64B.

1 Task 2A: Software Prefetching

1.1 Implementation

Cache blocking on M/N (block sizes, why panels then fit L1/L2), then `_mm_prefetch` hints ahead of use. State prefetch distance and locality hint chosen, and the `_builtin_prefetch` vs `_mm_prefetch` choice.

1.2 Baseline Profiling (perf)

`perf stat` for L1D / L2 / LLC misses and `sw_prefetch_access` across matrix sizes and prefetch parameters, for naive vs prefetch.

Config	L1D misses	L2 misses	LLC misses	sw_prefetch
naive	—	—	—	—
prefetch	—	—	—	—

Table 1: Cache metrics, naive vs prefetch. **TODO: fill values.**

1.3 Speedup vs. Matrix Size

plot placeholder: task2_prefetch_speedup_vs_size.png

Figure 1: Speedup of `matmul_prefetch` over `matmul_naive` vs. matrix size (cache-resident through beyond-LLC). **TODO: add plot.**

1.4 Speedup vs. Prefetch Distance

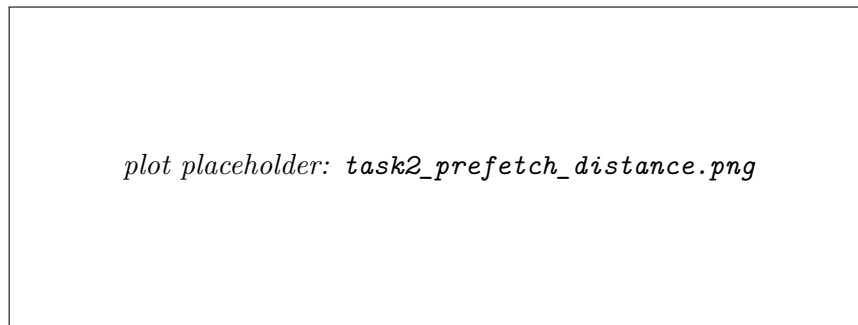


Figure 2: Speedup vs. prefetch distance (in floats / cache lines) for `matmul_prefetch`. **TODO: add plot.**

1.5 Speedup vs. Prefetch Locality Hint

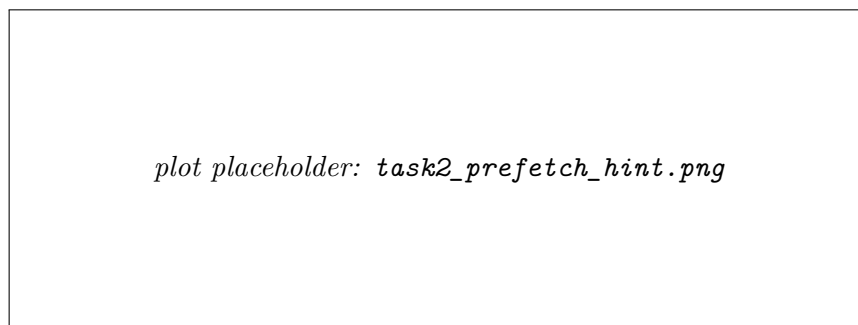


Figure 3: Speedup for `_MM_HINT_T0` / `T1` / `T2` / `NTA` locality hints. **TODO: add plot.**

1.6 Effect of Hardware Prefetchers

Enable/disable hardware prefetchers (MSR 0x1A4) and report the impact on the SGEMM speedup and on the software-prefetch benefit.

1.7 Questions

1. Trend in speedup with matrix size:

2. Best prefetch distance:

3. Cache fill level (locality hint) giving maximum speedup:

4. L1D / L2 / LLC miss trends across sizes and parameters:

5. Effect of hardware prefetchers:

2 Task 2B: SIMD (AVX2)

2.1 Implementation

The kernel keeps the natural $i \rightarrow j \rightarrow p$ loop order but register-blocks the j (column-of- B) loop by `step = 4`: each iteration of the inner loop computes four output elements $C[i][j..j+3]$ at once. Four `_mm256` accumulators (`acc0..acc3`) are held in registers and zeroed at the top of the block. The p loop then advances 8 floats at a time: one `_mm256_loadu_ps` pulls a lane of $A[i][p..p+7]$, which is reused across all four columns, and four `_mm256_fmadd_ps` multiply it by the four B -row lanes and add into the accumulators. This 1×4 register block amortizes the single A load and the loop overhead over four FMAs, and the four independent accumulator chains hide the FMA latency.

After the vector loop, each accumulator is reduced to a scalar by `hsum256` (an `extractf128 + add` to fold the two 128-bit halves, then two `hadd_ps`). Remainders are handled by scalar cleanup: a `p < K` tail loop finishes the last $K \bmod 8$ elements into the same four sums, and a `j < N` tail loop handles the last $N \bmod 4$ columns one at a time via the scalar-tail helper `dot_simd`. All loads are unaligned (`loadu`) so arbitrary `lda/lb/lc` are safe.

2.2 Baseline Profiling / Instruction Count

Retired user-space instructions were attributed to each kernel with `perf record -e instructions:u` followed by `perf report`: the per-call figure is the run's total event count times the kernel's overhead percentage, divided by the number of times the harness invokes that kernel in a single-stage run (naive: 1 reference + 1 warmup + 1 + 3 timed = 6; SIMD: 1 warmup + 1 + 3 = 5). Table 2 gives the 1024^3 point; Fig. 4 shows the full size sweep.

The reduction is essentially size-independent, $\approx 11.8\times$ for the 128-bit implementation and $\approx 23\times$ for the 256-bit implementation across the whole sweep, because it comes from the code's structure, not from cache behaviour. It exceeds the raw lane counts (4 and 8) because each vector iteration also folds a multiply and an add into one FMA and amortizes one A load, the loop-counter arithmetic and the branch over four columns of output. Doubling the vector width from 128 to 256 bits roughly halves the count, as expected.

Kernel	Instr./call (1024^3)	Reduction vs naive
<code>matmul_naive</code>	6.45×10^9	1.00×
<code>matmul_simd</code> (128-bit)	5.45×10^8	11.8×
<code>matmul_simd</code> (256-bit)	2.79×10^8	23.1×

Table 2: Retired user-space instructions per kernel call at $M=N=K=1024$, from `perf record/perf report`.

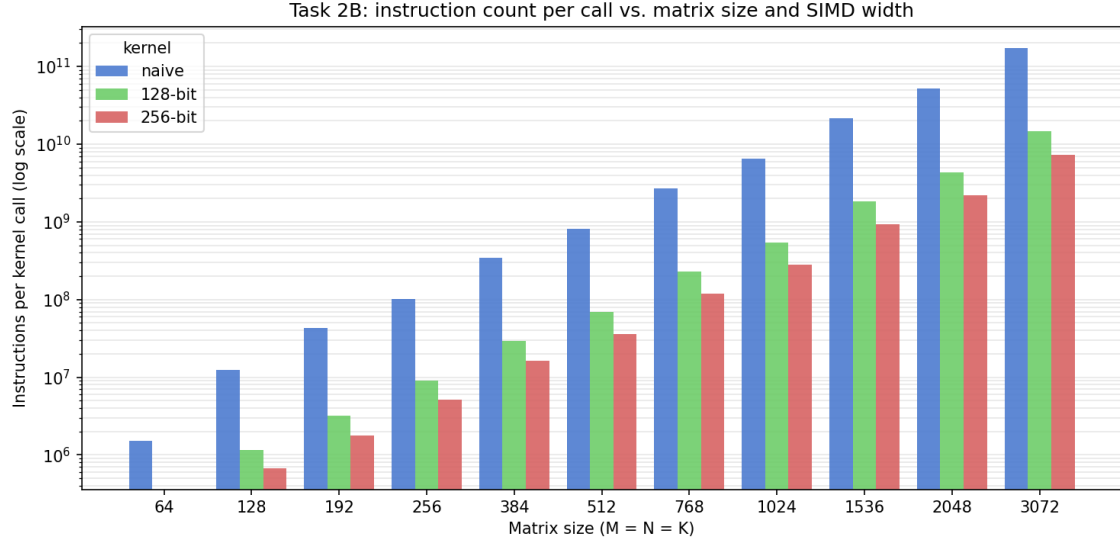


Figure 4: Instructions executed per kernel call vs. matrix size ($M=N=K$), for `matmul_naive`, the 128-bit SIMD kernel, and the 256-bit (AVX2) kernel. Per-call figures are the per-symbol instruction count from `perf record (event_count × overhead %)` divided by the number of times the harness invokes each kernel in a single-stage run (naive: 6, SIMD: 5). Log scale.

The 128-bit and 256-bit SIMD bars are absent at $N=64$: that workload runs in a few microseconds and `perf` collected too few samples (single-digit) to attribute a stable per-symbol percentage, so those points are dropped rather than shown as noise.

2.3 Performance

Stage	Time (ms)	GFLOP/s	Speedup
<code>matmul_naive</code> (ref)	496.506	4.33	1.00×
<code>matmul_simd</code> (256-bit)	92.051	23.33	5.39×

Table 3: `./bin/matmul simd 1024 1024 1024 1234`. At smaller, cache-resident sizes the SIMD speedup is much higher (up to 19× at $N=256$); see Fig. 5.

2.4 Speedup vs. SIMD Width

The final `src/matmul_simd.cpp` is the 256-bit SIMD implementation. Both implementations were swept over $M=N=K \in \{64, \dots, 3072\}$ against `matmul_naive`; the speedups are plotted in Fig. 5 and the absolute runtimes in Fig. 6. Below $N \approx 512$ the problem is cache-resident and the wider register roughly doubles the speedup (19× vs 8× at $N=256$); beyond the L2/L3 capacity both kernels become memory-bound and converge to ≈ 4 –5×, where vector width no longer matters.

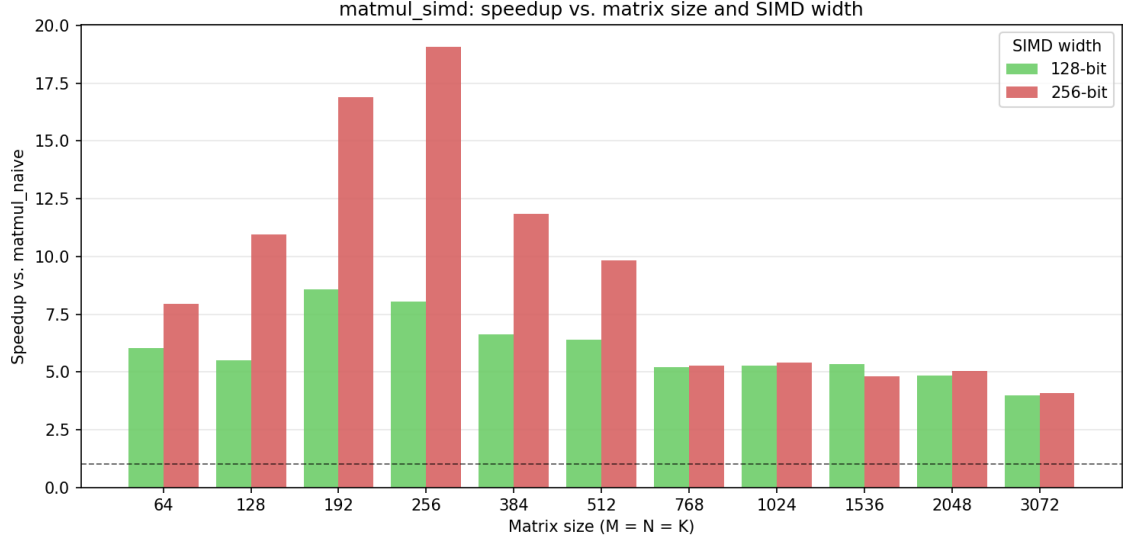


Figure 5: Speedup of `matmul_simd` over `matmul_naive` vs. matrix size ($M=N=K$), for the 128-bit and 256-bit SIMD kernels.

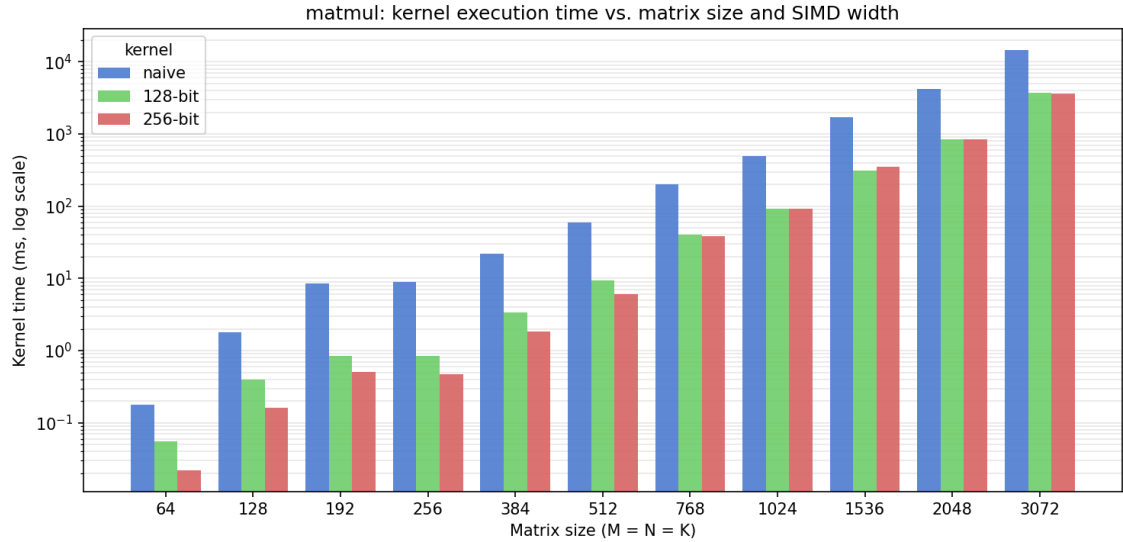


Figure 6: Kernel execution time (ms, log scale) vs. matrix size ($M=N=K$). Same runs as Fig. 5.

SIMD width	Metric	Matrix size N ($M=N=K$)										
		64	128	192	256	384	512	768	1024	1536	2048	3072
No SIMD (<code>matmul_naive</code>)	Instructions/call	1.52×10^6	1.25×10^7	4.31×10^7	1.02×10^8	3.41×10^8	8.08×10^8	2.72×10^9	6.45×10^9	2.18×10^{10}	5.16×10^{10}	1.74×10^{11}
	Time (ms)	0.177	1.778	8.598	8.934	21.990	59.789	201.384	496.506	1707.046	4249.303	14687.692
128-bit (SSE)	Instructions/call	—	1.15×10^6	3.20×10^6	9.07×10^6	2.92×10^7	6.93×10^7	2.31×10^8	5.45×10^8	1.83×10^9	4.32×10^9	1.46×10^{10}
	Time (ms)	0.056	0.395	0.844	0.843	3.362	9.502	40.829	93.164	312.891	854.104	3699.584
	Speedup vs. No SIMD	6.03×	5.51×	8.57×	8.04×	6.61×	6.38×	5.20×	5.26×	5.33×	4.84×	3.98×
256-bit (AVX2)	Instructions/call	—	6.73×10^5	1.77×10^6	5.17×10^6	1.62×10^7	3.61×10^7	1.19×10^8	2.79×10^8	9.30×10^8	2.19×10^9	7.34×10^9
	Time (ms)	0.022	0.162	0.510	0.468	1.858	6.075	38.203	92.051	354.612	843.911	3590.953
	Speedup vs. No SIMD	7.95×	10.94×	16.88×	19.08×	11.84×	9.84×	5.27×	5.39×	4.81×	5.04×	4.09×

Table 4: Instructions/call, kernel time, and speedup vs. matrix size, for each SIMD width. Instructions/call are retired user-space instructions from `perf record/perf report` (per-call = event count $\times$ overhead% divided by harness call count); times are the code’s median function-only `time(ms)`, not whole process `perf stat` time. Same source data as Table 2, Fig. 4, Fig. 5, and Fig. 6.

2.5 Questions

1. **Instruction count change (naive $\rightarrow$ SIMD):** Retired instructions per function call drop by about $11.8\times$ for the 128-bit implementation and $23\times$ for the 256-bit implementation, roughly constant across matrix size (Table 2, Fig. 4). This exceeds the raw lane counts (4 and 8) for two reasons: each `fmadd` replaces a separate multiply and add, and the 1×4 register block issues one A load plus one set of loop-counter/branch instructions per four FMAs instead of per one. The count halves cleanly from 128 to 256 bits because the work per instruction doubles while the loop structure is identical.
2. **Trends across matrix size $\times$ SIMD width; which width gives max speedup:** For cache-resident sizes ($N \lesssim 512$) speedup rises with N and scales with vector width, peaking at $N=256$: $19.1\times$ for 256-bit and $8.0\times$ for 128-bit. Here the code is compute-bound, so halving the instruction count with a wider register directly halves the runtime. As N grows past the L2/L3 capacity the $N \times K$ floats no longer fits and the function streams it from memory on every row of A ; both widths become memory-bound and converge to $\approx 4\text{--}5\times$, where SIMD width stops mattering. So 256-bit gives the maximum speedup at every size, but its advantage over 128-bit shrinks from $\sim 2\times$ to negligible as the workload leaves cache.

3 Task 2C: Software Prefetching + SIMD (Combined / Optimized)

3.1 Implementation

Everything combined: loop reordering, register blocking + unrolling, AVX2 micro-kernel, cache tiling, and software prefetch. Give the final block sizes, micro-kernel shape, prefetch distance/hint, and the tuning process. Note arbitrary $M/N/K/lda/ldb/l dc$ handling (scalar tails) since this kernel is injected into llama.cpp.

3.2 Combinations Evaluated

- **TODO:** SIMD only — ...
- **TODO:** SIMD + tiling — ...
- **TODO:** SIMD + tiling + prefetch — ...
- **TODO:** SIMD + tiling + prefetch + unroll (chosen) — ...

3.3 Final Performance Summary

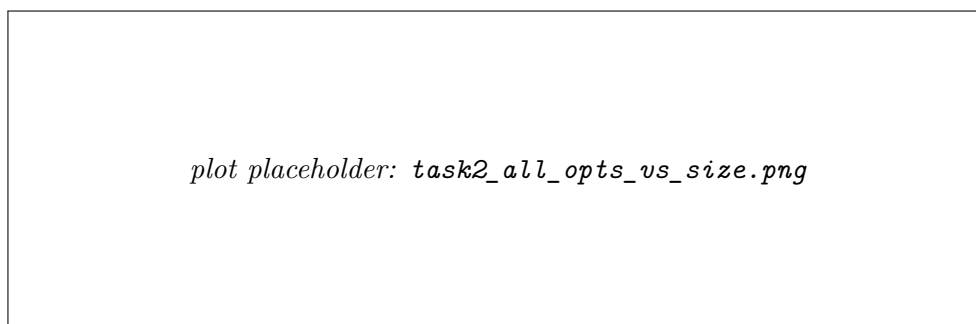


Figure 7: Speedup of `matmul_simd`, `matmul_prefetch`, and `matmul_optimized` over `matmul_naive` vs. matrix size ($M=N=K$, 128 through 2048+). **TODO: add plot.**

Stage	Speedup @ 1024 ³	Speedup @ 2048 ³
<code>matmul_simd</code>	—	—
<code>matmul_prefetch</code>	—	—
<code>matmul_optimized</code>	—	—

Table 5: Speedup of each stage. **TODO: fill values.**

3.4 Harness Score

Paste the graded table and TOTAL from a full `./bin/matmul` run; state which tier `matmul_optimized` reaches.

3.5 Discussion

Why SIMD before tiling (compute-bound $\rightarrow$ memory-bound); why cache blocking becomes decisive once the kernel is fast; why prefetch adds on top; why the speedup varies with matrix size relative to L1/L2/LLC capacity.

4 llama.cpp Integration

4.1 Setup

Pinned tag `b5731`, `stories260K.gguf` (F32), single-thread, injection path (`inject.py` patches `ggml_compute_forward_mul_mat`; `ggml_student_sgemv.cpp` adapts the kernel; A/B swap via dot-product symmetry).

4.2 Results

Metric	llama.cpp native	student kernel
Kernel used marker seen	—	—
Output matches stock (token-for-token)	—	—
Prompt-eval tokens/sec	—	—
Generation tokens/sec	—	—

Table 6: make llama-demo comparison. **TODO: fill values.**

4.3 Discussion

Compare prompt-eval (batched, fast path) vs generation ($M=1$, scalar tail). Why the student kernel does / does not beat ggml's native matmul.